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Daniel Smith

Los Angeles, CA (U.S. Citizen)

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Incoming UCLA M.S. Computer Science student with work experience in Big Data and Machine Learning. Currently seeking opportunities to apply skills for the Summer of 2022 in Data Science or Software Engineering.

Skills

Tools and Languages **Advanced:** Python, C/C++, Java, R, MySQL, Linux, Vim, VS Code, Git, NumPy, Pandas, PyTorch
Proficient Docker, Django, MongoDB, JavaScript, React, Node.js, ElasticSearch, Google Cloud Platform

Communication English(Native), Korean(Fluent)

Experience

Data Science Intern **June 2021 — Sept. 2021**
Lockheed Martin *El Segundo, CA*

- Developed frontend and backend of an Internal Django Website with one other intern, in order to aggregate Internal Documents, in addition to implementing search functions using ElasticSearch deployed in Docker.
- Created Pipeline and Python script to use and export data from MongoDB in order to automate a custom A/B Testing Script for search recommendations.
- Retrained Google's Word2Vec model with Senior AI Staff Scientist, Mathew Horak, to recognize company's keywords in searches aggregated from MongoDB, as well as implement K-Means Clustering to quantify the amount of search sessions
- Led intern team of 5 in business proposal to use proprietary ML software for object recognition in satellite images and presented it to Lockheed Martin Fellow, Sada Narayanappa.

Bioinformatics Undergraduate Researcher **Mar. 2020 — June 2021**
Scalable Analytics Institute *Los Angeles, CA*

- Worked under Professor Wei Wang and PhD candidate, Guangyu Zhou, to implement the PSIBLAST Algorithm on the UniRef90 Database, and augment the select features to create a PSSM (Position Specific Scoring Matrix)
- Developed LSTM model in PyTorch that reads in protein sequences as well as a PSSM and classifies anomalies in resulting genome sequences 6% more accurately than conventional models

Hackathon Organizer and Mentor **June 2020 — Oct. 2020**
UCLA Transfer Bridge *Los Angeles, CA*

- Administered and led workshops on Python and Data Structures in C++ to prepare transfer students with the skills and knowledge necessary to succeed at UCLA
- Responsible for managing logistics and budget of $5000 for an online Electric Car Hackathon with Electrical and Computer Engineering Professor, Gregory Pottie, as well as 3 Graduate Students for 60 incoming Transfer Students

Projects

Pix2PixGAN Clone

- Created a clone from scratch of the Pix2PixGAN in PyTorch consisting of a PixelGAN generator with Up-sampling and Down-sampling Modules as well as a U-Net discriminator *(https://tinyurl.com/73wtyrv7)*
- Developed Pipeline for aggregating and converting hand drawn images to grayscale in order to train Pix2PixGAN to generate colored versions of the image *(https://colab.research.google.com/drive/1JIn-osvs2utb42cYKI5dPkDVea3l3RQA?usp=sharing)*

Bruin Web Server

- Developed backend for web server with 3 other developers in C++, using the Boost library and hosted it on Google Cloud Platform. Presented product to Google Staff Software Engineer, Michael Burns, and Senior Software Engineer, Eric Hennigan
- Implemented CRUD API capable of using POST, PUT, GET, and DELETE functions. Added markdown support using a third party library, Maddy, capable of parsing input from an md file and rendering it onto an html webpage

Education

University of California, Los Angeles
Master of Science - Computer Science *Concentration: AI/ML* *Expected Graduation:* **Spring 2023**
Bachelor of Science - Computer Science *GPA: 3.736/4.000* **Sept 2019 — June 2022**

Relevant Courses: Deep Learning for Computer Vision, Applications of NLP, Probability Theory, Mathematical Statistics, Data Science Fundamentals, AI Fundamentals, Data Structures & Algorithms, Theory of Computing

Extracurriculars and Awards

UCLA Engineering Transfer Center - *Engineering Resource Center*: Data Structures in C++ Tutor **Oct. 2020 — Present**
UCLA Bruin Resource Center - *Former Foster & Emancipated Youth Organization*: Bruin Guardian Scholar **Sept. 2019 — Present**
UCLA Transfer Bridge Hackathon Winner - *Awarded $1600 for 1st place in code and functionality of a Robotic Crane* **June 2019**